UNIVERSITY OF PENNSYLVANIA ESE 5700

The 6T SRAM bitcell

Project 1, Part 1: design, analysis and present evidence

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Cadence GPDK 18 nm (`cds_ff_mpt`) $V_{DD} = 0.8$ V $\beta = 2, \gamma = 1$ cell area — μm^2 = — F^2

Executive summary

The symmetric 6T cell was sized by hand analysis and a three-point schematic β sweep: at 0.8 V, raising β from 1 to 2 increases read SNM from 126.6 to 156.3 mV and lowers the read-disturbed zero node from 134.3 to 77.1 mV, so the $\beta = 2$, $\gamma = 1$ cell (PD 4 fins, PU 2 fins, PG 2 fins, $L = 18$ nm) was selected. The cell was laid out with a horizontal wordline and a vertical bitline pair so that mirrored neighbours share bitline contacts and supply rails, and signed off with the course Pegasus DRC and LVS decks (Table 3: DRC —; LVS —). The cell occupies — μm^2 = — F^2, giving a density factor D = — for the Part 2/3 area accounting. From the extracted view, one cell presents C_{BL} = — aF and C_{WL} = — aF; at 0.8 V its hold SNM is — mV, read SNM — mV, write margin — V and read current — μA, and the supply sweep in Table 6 states whether it still reads and writes at 0.5 V. Entries shown as "—" are not yet measured; nothing in this report is reported without a run behind it.

Table 1: What this report can establish now. Values are from `results/part1/read-snm/beta*/snm.json`.

β	PD / PG / PU fins	Schematic read SNM (mV)	Read-disturbed zero node (mV)
1.0	2 / 2 / 2	126.6	134.3
1.5	3 / 2 / 2	145.8	96.6
2.0	**4 / 2 / 2**	**156.3**	**77.1**

1 Assignment scope and evidence standard

The current handout requires: sized symmetric schematic and hand analysis; a three-point schematic β sweep; custom tileable layout with area and DRC/LVS summaries; per-cell BL and WL capacitance after extraction; post-layout hold/read SNM, write margin, cell current and supply dependence; a transient write/read sequence; and a discussion of the limiting trade-off. It also requires individual team contributions. Figures and numbers below are labeled as *measured schematic*, *hand estimate*, or *open*. The supplied handout sets $F = 10$ nm for area in F^2, and supply points of 0.8, 0.7 and 0.5 V.

2 Cell schematic and selected dimensions

A 6T cell consists of cross-coupled inverters INV1 and INV2 and two pass-gates. Both storage nodes connect to precharged bitlines only while WL is high. For the candidate schematic, the captured device settings are $L = 18$ nm and $n_{\text{fin}} = 4$ for each pull-down, $n_{\text{fin}} = 2$ for each pull-up, and $n_{\text{fin}} = 2$ for each pass-gate. Thus the nominal fin-count ratios are

$$\beta = \frac{(W/L)_{\text{PD}}}{(W/L)_{\text{PG}}} \simeq \frac{4}{2} = 2, \qquad \gamma = \frac{(W/L)_{\text{PU}}}{(W/L)_{\text{PG}}} \simeq \frac{2}{2} = 1. \tag{1}$$

The effective process-model strengths are not simply fin counts; the ratios express the intended relative sizing. Both halves are identical, meeting the symmetric-cell requirement at schematic level.

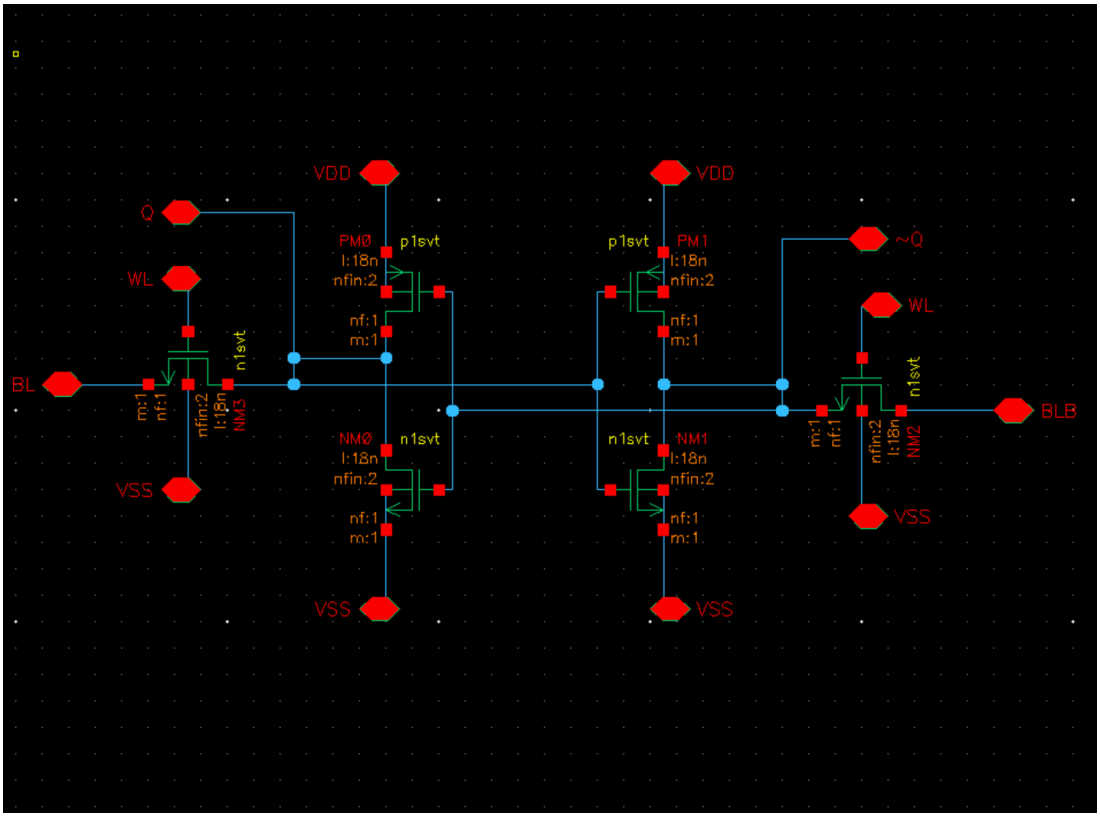


Figure 1: Original Virtuoso cell schematic, with device fin counts and 18 nm length visible. The plot and bench evidence in later sections comes from this family of schematics.

3 Hand sizing analysis

3.1 Read disturb and the cell ratio

Assume $Q = 0$, $\overline{Q} \approx V_{DD}$, WL= V_{DD} and BL=BLB= V_{DD}. Current flows from BL through the pass-gate into Q and down through the pull-down. The handwritten circuit and divider in Figure 2 were part of the original analysis. They identify the two opposing devices and the node whose rise can upset INV2.

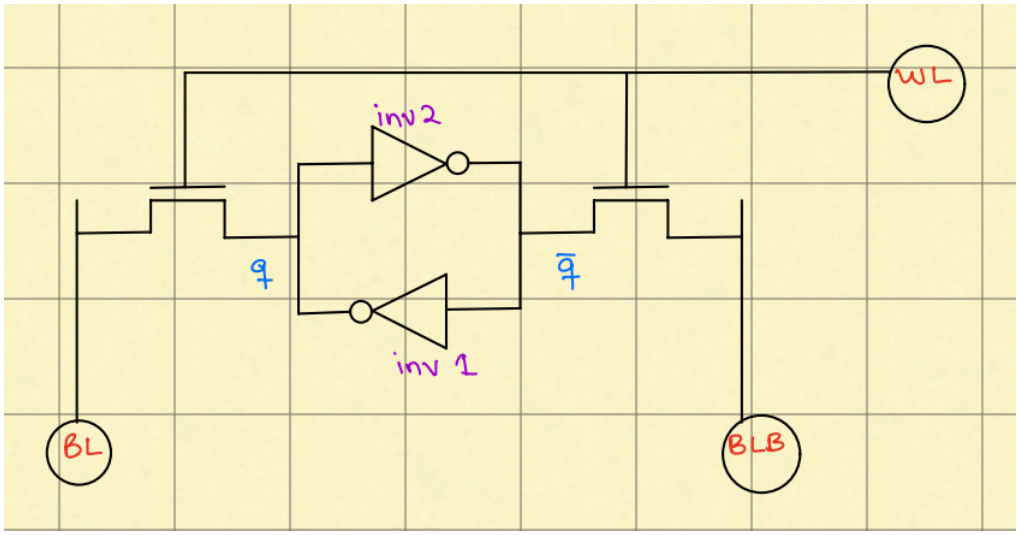


Figure 2: Original handwritten read-state diagram, with node labels.

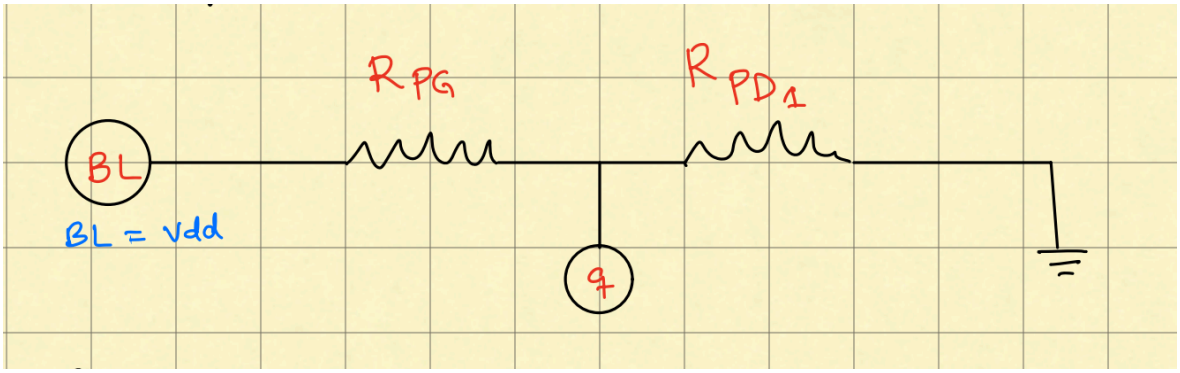


Figure 3: Original handwritten read-divider sketch.

For equal nominal channel length, a resistive first estimate gives

$$\frac{V_Q}{V_{DD}} = \frac{R_{\mathrm{PD}}}{R_{\mathrm{PG}} + R_{\mathrm{PD}}} \approx \frac{1}{1+\beta}, \qquad V_Q \approx \frac{V_{DD}}{1+\beta}. \tag{2}$$

At 0.8 V, this simple expression predicts 0.400, 0.320 and 0.267 V for $\beta = 1$, 1.5 and 2. These are *hand estimates*, much higher than the DC crossings in Table 1 because the devices are nonlinear and the ratio is only a proxy for their actual strengths. Their useful prediction is the downward trend with increasing β.

For a more physical but still ideal long-channel model, set $E = V_{DD} - V_{Tn}$ and $k = \mu_n C_{ox}(W/L)$. The pass-gate is saturated with $V_{GS} = V_{DS} = V_{DD} - V_Q$; the pull-down is in triode if $V_Q < E$. Neglecting leakage and body effect, Kirchhoff's law at Q gives

$$\underbrace{\tfrac{1}{2}k_{\mathrm{PG}}(E - V_Q)^2}_{I_{\mathrm{PG}}} = \underbrace{k_{\mathrm{PD}}(EV_Q - \tfrac{1}{2}V_Q^2)}_{I_{\mathrm{PD}}}. \tag{3}$$

Solving the physical root using $k_{\mathrm{PD}}/k_{\mathrm{PG}} \approx \beta$ yields

$$V_Q = E\left(1 - \sqrt{\frac{\beta}{1+\beta}}\right). \tag{4}$$

The positive quadratic root violates the assumed operating region. The original handwritten steps in Figure 4 show the pass-gate current, current-ratio equation and closed form; the equations above state them legibly. Neither model substitutes for the GPDK simulation.

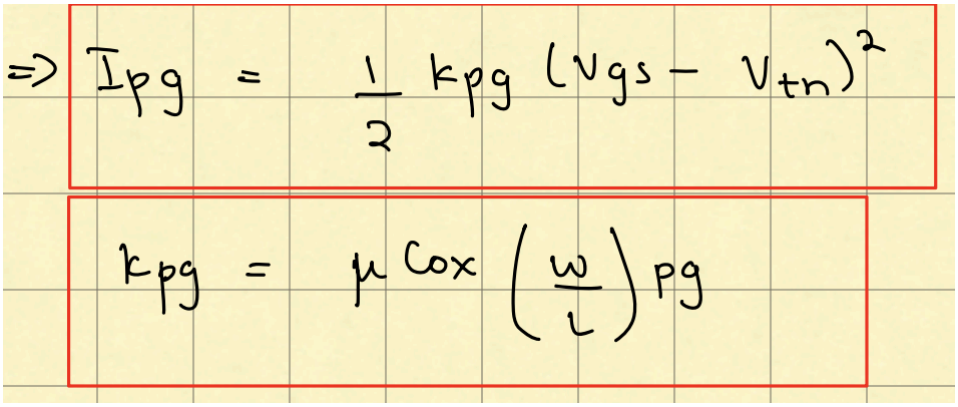


Figure 4: Original handwritten pass-gate current equation.

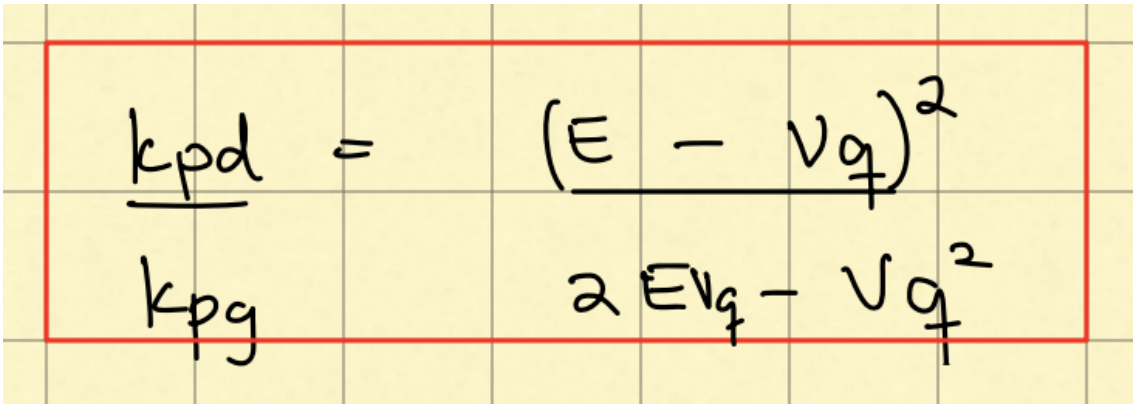


Figure 5: Original handwritten PD/PG current balance.

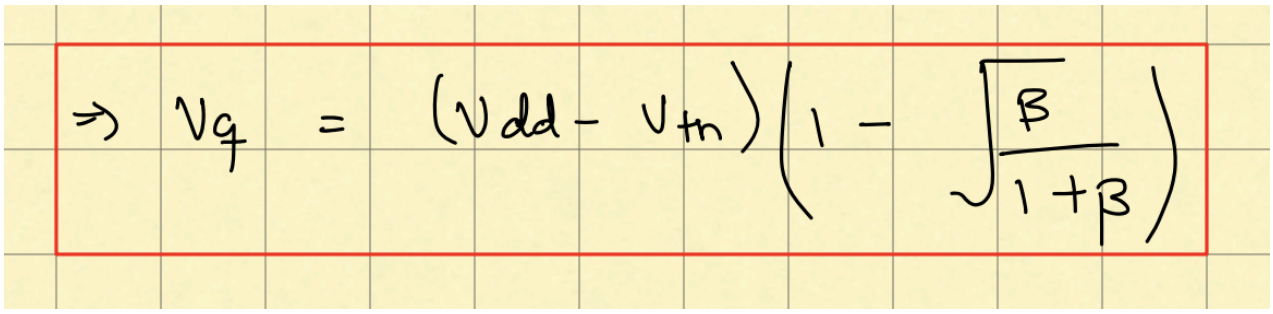


Figure 6: Original handwritten read-disturb voltage root.

The disturbed node drives INV2. The read criterion is $V_Q < V_M$, where V_M is INV2's switching voltage. In a square-law inverter approximation,

$$V_M = \frac{V_{Tn} + B(V_{DD} - |V_{Tp}|)}{1 + B}, \quad B = \sqrt{\frac{k_{\mathrm{PU}}}{k_{\mathrm{PD}}}} \approx \sqrt{\frac{\gamma\mu_p}{\beta\mu_n}}. \tag{5}$$

V_M is not independently measured here. Increasing β lowers the disturb voltage, but it also changes V_M, area and later bitline loading; measured read SNM is therefore the deciding schematic quantity.

3.2 Write contention and the pull-up ratio

For a write of zero into a stored one, take $Q \approx V_{DD}$ initially, $\overline{Q} \approx 0$, BL= 0, WL= V_{DD}. The pass-gate discharges Q against an ON pull-up. The first-order divider reverses the read case:

$$\frac{V_Q}{V_{DD}} \approx \frac{R_{\mathrm{PG}}}{R_{\mathrm{PG}} + R_{\mathrm{PU}}} \approx \frac{\mu_p\gamma}{\mu_n + \mu_p\gamma}. \tag{6}$$

A smaller γ weakens the pull-up relative to the pass-gate and lowers the written node. The receiving inverter must switch before the cell can settle back, so the condition is $V_Q < V_M$. If $\rho = \mu_n/\mu_p$, the divider implies $\gamma < \rho V_M/(V_{DD} - V_M)$, conditional on the assumed resistor model. Without a measured V_M or process strengths this is a design inequality, not a numeric proof that $\gamma = 1$ writes.

The more detailed hand current balance treats the pass-gate as triode and the pull-up as saturated while $V_Q < |V_{Tp}|$. With $E = V_{DD} - V_{Tn}$ and $F = V_{DD} - |V_{Tp}|$,

$$I_{\mathrm{PG}} = k_{\mathrm{PG}}(EV_Q - V_Q^2/2), \tag{7}$$

$$I_{\mathrm{PU}} = \tfrac{1}{2}k_{\mathrm{PU}}F^2, \qquad \frac{k_{\mathrm{PU}}}{k_{\mathrm{PG}}} \approx \gamma\frac{\mu_p}{\mu_n}, \tag{8}$$

$$V_Q = E - \sqrt{E^2 - \gamma(\mu_p/\mu_n)F^2}. \tag{9}$$

The other quadratic root is inconsistent with the assumed pass-gate region. Equation (9) also ceases to apply if the pull-up enters triode; no extracted model parameters have been inserted to make a fabricated numeric write limit. A stronger pass-gate helps writing but reduces β and adds load; the $\gamma = 1$ candidate needs a write-margin sweep before acceptance.

4 Read-SNM bench and reduction method

4.1 Circuit setup

The available Cadence benches break the inverter feedback for DC transfer curves. The Q sweep records $\overline{Q}$; the complementary sweep records Q. Both use WL= V_{DD}, BL=BLB= V_{DD} and a 0–0.8 V sweep. Figure 7 identifies the actual circuit setup rather than replacing it with a generic butterfly drawing.

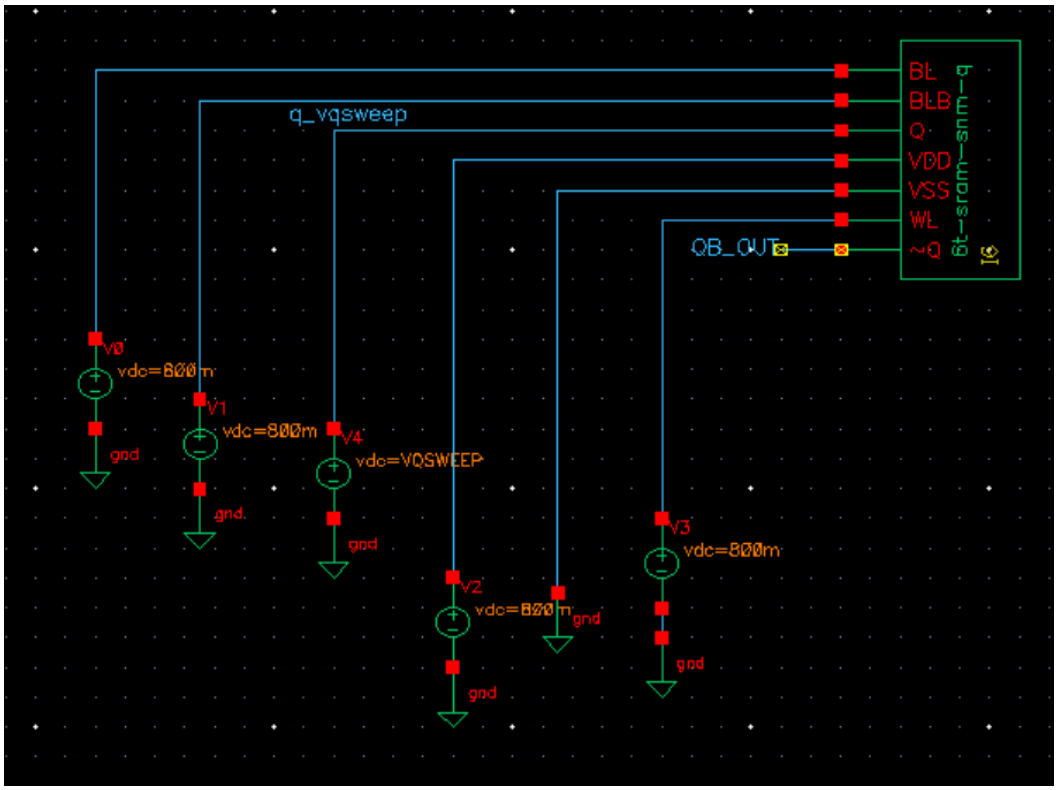


Figure 7: Original Virtuoso Q-side read-SNM sweep bench.

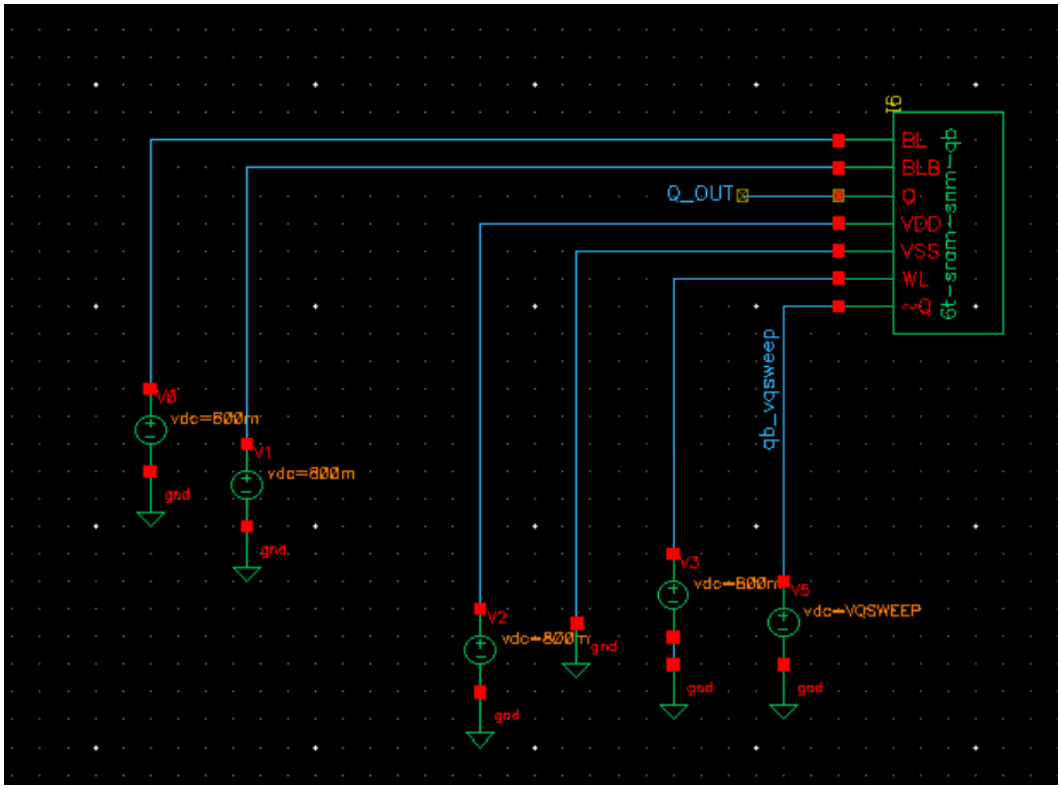


Figure 8: Original Virtuoso $\overline{Q}$-side read-SNM sweep bench.

4.2 From ViVA export to a margin

Each run has a four-column `raw.csv` ViVA export. `scripts/check_snm_sweep.py` detects which trace is normal and which is reflected, writes normalized `q.csv` and `qb.csv`, mirrors one VTC about $y = x$, locates the three intersections, and fits the largest axis-aligned square within each of the two lobes. The reported margin is

$$\mathrm{SNM}_{\mathrm{read}} = \min(\mathrm{SNM}_L, \mathrm{SNM}_R). \tag{10}$$

A 16,001-point voltage grid is used for the final result. Repeating the fit at 4,001 points changes the side length by at most 0.05 mV. The square coordinates and crossings are retained in each `snm.json`. The low-voltage crossing is reported as the DC "0"-node read-disturb operating point. This is a *schematic* analysis, even though Part 1 ultimately asks for the same measurements after extraction.

5 Measured schematic read stability

Table 2: Three-point read sweep at 0.8 V, from the raw Spectre exports. Extra digits preserve calculation traceability.

β	PD fins	SNM_L (mV)	SNM_R (mV)	$\mathrm{SNM}_{\mathrm{read}}$ (mV)	low crossing (mV)
1.0	2	126.6	126.6	126.6	134.3
1.5	3	145.8	145.8	145.8	96.6
2.0	4	156.3	156.3	156.3	77.1

For submission at the handout's requested two significant figures, the three read-SNM results are 0.13, 0.15 and 0.16 V; the corresponding disturbed zero-node crossings are 0.13, 0.097 and 0.077 V. The additional digits in Table 2 are retained to distinguish the underlying calculations. The selected candidate gains 29.7 mV (23%) over $\beta = 1$ at this schematic operating point. The low crossing drops by 57.2 mV. That direction agrees with Equations (2) and (4). No layout-area penalty or write-margin trade-off has been measured yet.

5.1 Original Cadence butterflies with fitted squares

Figures 9–11 are the *original ViVA screenshots*; the cyan outlines are report overlays placed at the coordinates saved by the maximum-square script. The voltage axes in ViVA use different horizontal and vertical pixel scales, so a square in volts appears rectangular on screen. The outlines are analytical annotations, not marks produced by Cadence. They make the two fitted lobes visible on the actual captured curves.

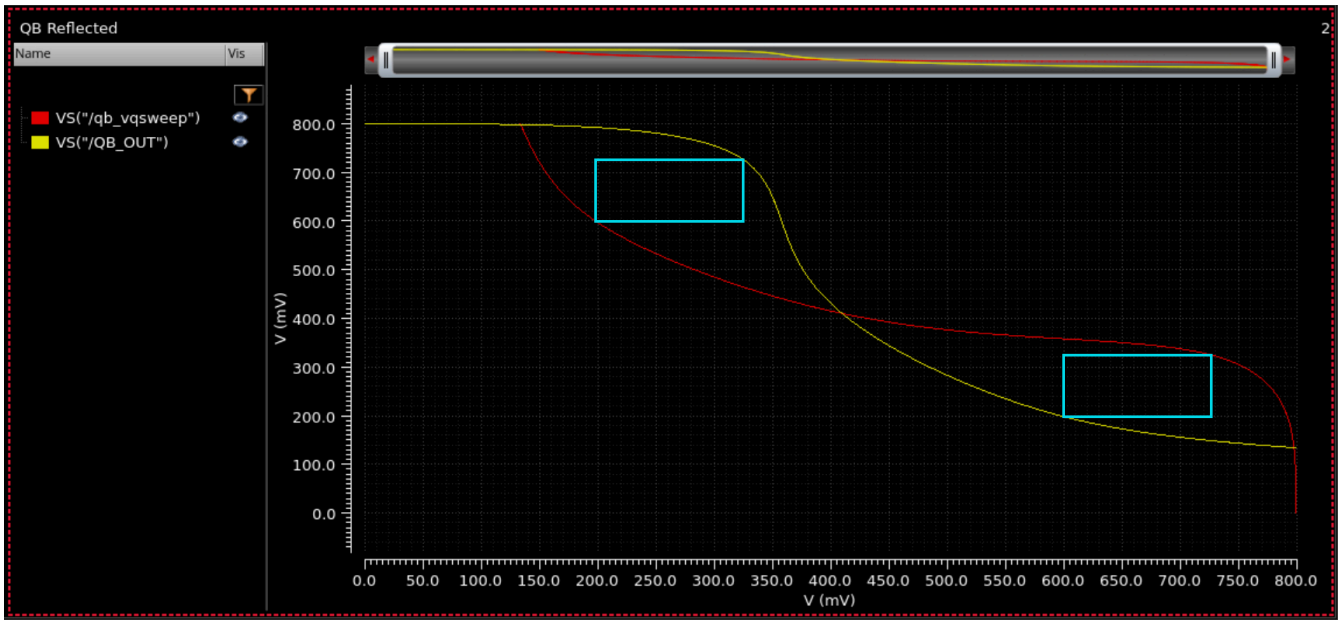


Figure 9: Original $\beta = 1$ ViVA butterfly with both computed largest-square locations overlaid; SNM = 126.6 mV.

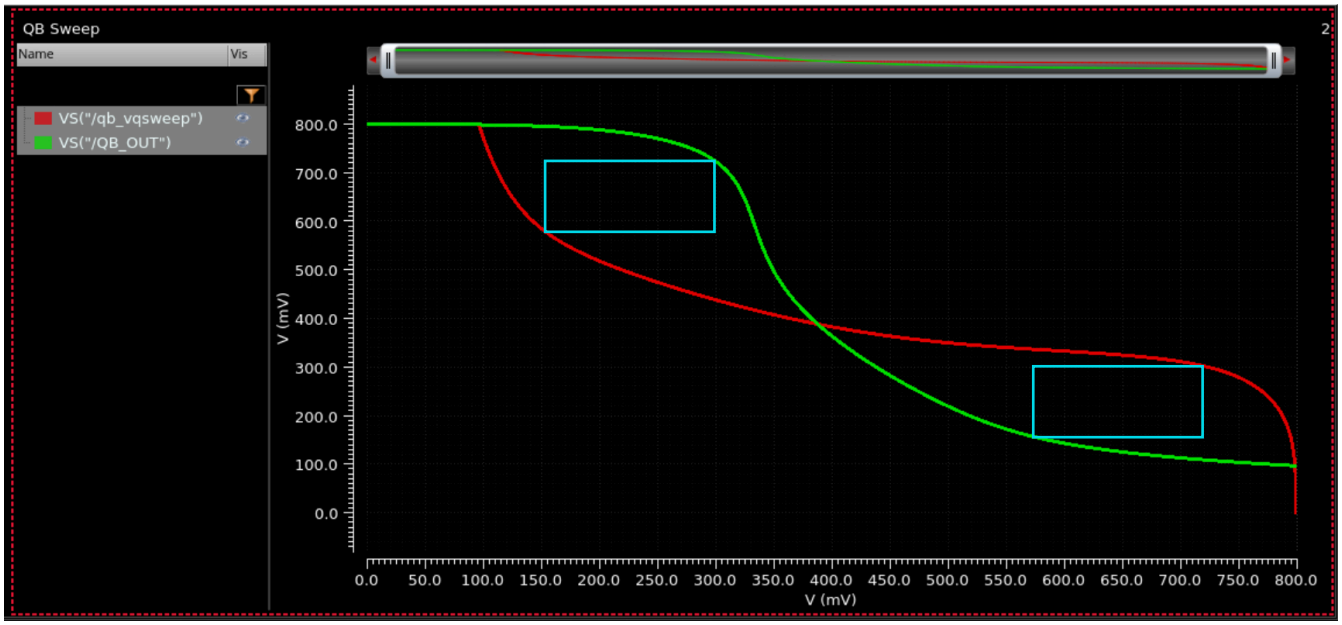


Figure 10: Original $\beta = 1.5$ ViVA butterfly with fitted-square overlays; SNM = 145.8 mV.

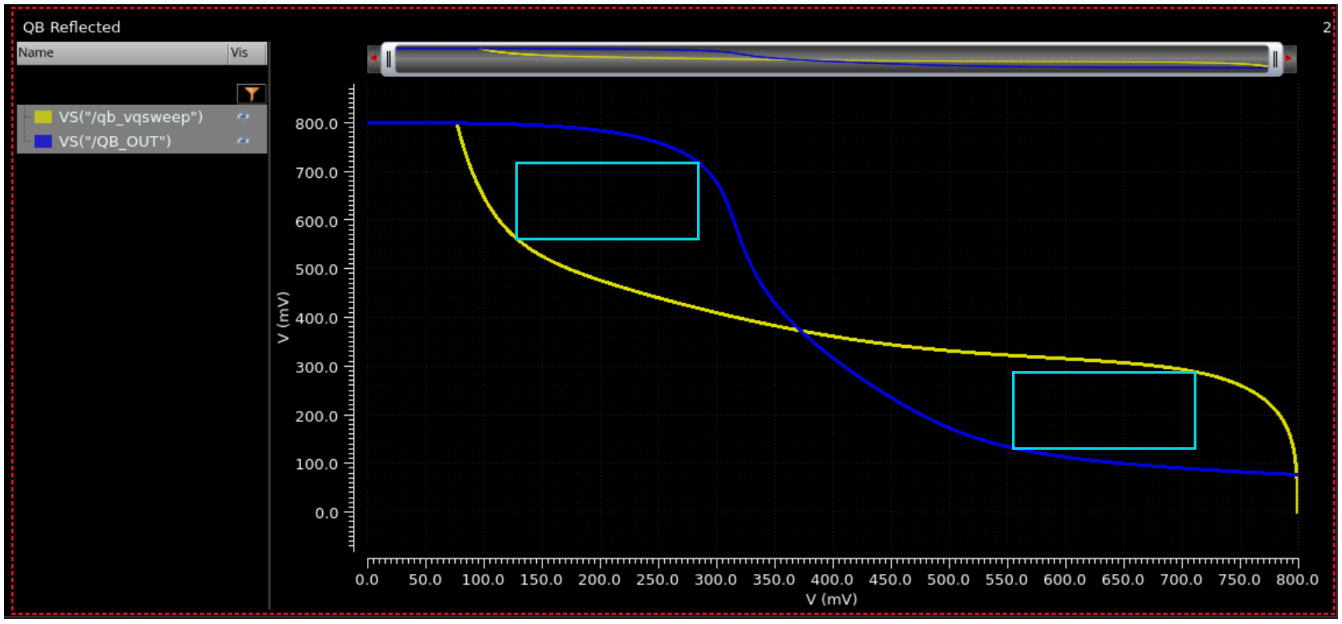


Figure 11: Original $\beta = 2$ ViVA butterfly with fitted-square overlays; SNM = 156.3 mV. This is the selected schematic candidate.

6 Layout (handout 1.2)

6.1 Floorplan and tiling intent

The cell is laid out in `proj1_lib/6t-sram_final_layout/layout` (technology `cds_ff_mpt`, Virtuoso IC23.1 Layout XL) against the connectivity reference `proj1_lib/6t-sram/schematic`. The six PDK FinFET P-cells are kept at their schematic settings: $L = 18$ nm, one finger, pull-ups PM0/PM1 at 2 fins, pull-downs NM0/NM1 at 4 fins, pass-gates NM3/NM2 at 2 fins. The floorplan follows the handout's tiling rule:

- the two pull-ups share one n-well strip under the top VDD rail; the pull-downs and pass-gates sit in the p-substrate region below it, so a vertically mirrored neighbour shares the rail and the well edge;
- the wordline runs *horizontally* across the full cell width and polysilicon gates of both pass-gates are tapped from it, so horizontally abutted cells form one continuous WL;
- BL and BLB run *vertically* on the metal level above WL at the left and right cell edges, over the pass-gate drain contacts, so a vertically mirrored neighbour shares each bitline contact;
- VSS is placed on the bottom edge and VDD on the top edge so that both rails are shared by mirrored rows; Q and $\overline{Q}$ stay internal.

Placement was generated from the schematic (Layout XL *Generate All From Source*, then *Place As In Schematic*) and the devices were then snapped to the PDK fin grid by hand; the signal nets were routed with the Virtuoso Space-based Router (*Route* → *Automatic Routing*) restricted to the local-interconnect and M1/M2 layers, and any router result that violated the restricted-direction LiAct or M1 end-of-line rules was redrawn manually. Verification used the course Pegasus 23.2 DRC and LVS decks installed under `proj1_lib/lvs`, not the preliminary *Verify* → *Process Rules* check.

6.2 Result

Final DRC/LVS-clean cell layout capture (all layers, PR boundary visible, WL horizontal, BL/BLB vertical).
`media/part1/6t_sram_layout_final.png` **Not yet evidenced**

Figure 12: Final bitcell layout, `proj1_lib/6t-sram_final_layout`. Width — μm (bitline pitch direction), height — μm (wordline pitch direction).

Table 3: Layout signoff and area. $F = 10$ nm per the handout, so $A_{F^2} = A_{\mu m^2}/(0.01\,\mu\text{m})^2 = 10^4 A_{\mu m^2}$.

Quantity	Value	Source
Pegasus DRC	—	`proj1_lib/drc` run summary (Fig. 13)
Pegasus LVS vs. `6t-sram/schematic`	—	`proj1_lib/lvs` run summary (Fig. 13)
Cell width × height	— × — μm	PR boundary of the layout
Cell area A_{cell}	— μm^2 = — F^2	width × height
Density factor D	—	Eq. (11)

Pegasus Results Viewer / log summaries showing the DRC violation count and the LVS comparison result.
`media/part1/pegasus_drc_lvs_summary.png` **Not yet evidenced**

Figure 13: DRC and LVS summaries for Figure 12.

The area in Table 3 is the PR-boundary area that abutting cells repeat at; it includes the half-rails and half-bitline-contacts that are shared with the mirrored neighbours, which is the area the Part 2 array actually tiles.

6.3 Density factor for the area accounting

The handout estimates un-laid-out blocks as $A_{\text{est}} = D\sum_i W_i L_i$ with $D = A_{\text{cell}}/\sum_{\text{cell}} W_i L_i$ measured from this cell. For FinFETs the drawn width is quantized, so we state the convention: one fin counts as the PDK fin pitch, $W_{\text{fin}} = 48$ nm (the `FinArea` grid of `cds_ff_mpt`), and $L = 18$ nm for every device. The cell has 4+4+2+2+2+2 = 16 fins, so

$$\sum_{\text{cell}} W_i L_i = 16 \times 0.048 \times 0.018\ \mu\text{m}^2 = 0.0138\ \mu\text{m}^2, \qquad D = \frac{A_{\text{cell}}}{0.0138\ \mu\text{m}^2} = - - -. \tag{11}$$

The same W_{fin} convention must be used for every $\sum W_i L_i$ in Parts 2 and 3 so that D cancels consistently.

7 Extraction (handout 1.3)

Parasitics were extracted from Figure 12 with the tutorial Quantus flow (C+CC, extracted view `6t-sram_final_layout/av_extracted`). All measurements in Section 8 and 9 use that view. The per-cell loads that Part 2 inherits are

Table 4: Extracted node capacitance of one cell (WL = 0, bitlines held at V_{DD} for the DC operating point).

Node	Capacitance	Meaning for Part 2
BL (one bitline, one cell)	— aF	$C_{BL} \approx 16\times$ this + wire per column
WL (one cell)	— aF	$C_{WL} \approx 4\times$ this + wire per row

The bitline number is dominated by the pass-gate drain junction plus the shared bitline contact; the wordline number by two pass-gate gate capacitances plus the WL wire across the cell. Fifteen copies of the BL value are the stand-in column load used in the functional check.

8 Post-layout stability characterization (handout 1.4)

All benches below use the extracted view at $V_{DD} = 0.8$ V unless stated; the same largest-square reduction as Section 4 (`scripts/check_snm_sweep.py`) is applied to the exported butterflies, and for any asymmetry SNM $= \min(\mathrm{SNM}_L, \mathrm{SNM}_R)$.

Table 5: Post-layout stability results at $V_{DD} = 0.8$ V. Setups are the handout's 1.4a–d.

Item	Bench	Result
Hold SNM	WL= 0; feedback broken; sweep V_Q, record $V_{\overline{Q}}$ and vice versa; overlay.	— mV
Read SNM	WL= V_{DD}, BL=BLB= V_{DD} (ideal sources); same butterfly.	— mV
Read "0" node	DC voltage of the disturbed zero node during the read butterfly; compare to Eq. (4) and the 77.1 mV schematic crossing.	— mV
Write margin	Store $Q = 1$; WL= V_{DD}; BLB held at V_{DD}; sweep BL from V_{DD} to 0; $\mathrm{WM}_{BL} = V_{\mathrm{trip}}$.	— V
Cell current	WL= V_{DD}, both bitlines at V_{DD}; current drawn from the zero-side bitline.	— μA

Hold butterfly (extracted, 0.8 V).
`media/part1/hold_snm_butterfly.png` **Not yet evidenced**

(a) Hold butterfly, $\mathrm{SNM}_{\mathrm{hold}}$ = — mV.

Read butterfly (extracted, 0.8 V).
`media/part1/read_snm_butterfly_postlayout.png` **Not yet evidenced**

(b) Read butterfly, $\mathrm{SNM}_{\mathrm{read}}$ = — mV.

Figure 14: Post-layout butterfly plots with fitted largest squares.

Write-margin sweep: Q and $\overline{Q}$ versus V_{BL} swept from V_{DD} down to 0 with BLB = V_{DD}, WL = V_{DD}.
`media/part1/write_margin_sweep.png` **Not yet evidenced**

Figure 15: Write-margin sweep; the cell flips at $V_{BL} = \mathrm{WM}_{BL}$ = — V.

8.1 Supply dependence (handout 1.4e)

Table 6: Read SNM and write margin versus supply, extracted view.

V_{DD} (V)	$\mathrm{SNM}_{\mathrm{read}}$ (mV)	WM_{BL} (V)	Comment
0.8	—	—	nominal
0.7	—	—	
0.5	—	—	—

At the schematic level the 0.8 V read SNM was 156.3 mV (Table 2); the extracted 0.8 V entry above is the number that supersedes it. Whether the cell still reads and writes at 0.5 V is answered by the last row: a positive write margin and a non-collapsed read butterfly are both required.

9 Functional check (handout 1.5)

The transient bench drives WL and the precharge devices with ideal pulses and loads each bitline with the extracted cell plus 15 copies of the per-cell bitline capacitance from Table 4, standing in for the 16-row Part 2 column. The sequence is write 0, read, write 1, read.

Transient waveforms WL, BL, BLB, Q, $\overline{Q}$ for write 0 / read / write 1 / read with 15 extra per-cell bitline loads.
`media/part1/functional_check.png` **Not yet evidenced**

Figure 16: Functional check. The reads must leave $Q/\overline{Q}$ undisturbed and both writes must complete within the WL pulse.

10 What limits the cell, and what changes for Part 2

Read stability is set by the pass-gate/pull-down divider: the schematic sweep moved the disturbed zero node from 134 mV ($\beta = 1$) to 77 mV ($\beta = 2$) and raised read SNM by 30 mV, and the extracted value in Table 5 is the one that must clear the Part 2 noise budget. The cost of $\beta = 2$ is the four-fin pull-down, which sets the cell height and most of C_{WL}, and the two-fin pass-gate, which sets C_{BL} and the read current that the Part 2 sense path has to integrate. Writability depends on $\gamma = 1$ and is only proven by the write-margin sweep; if WM_{BL} at 0.5 V is marginal, the remedy for Part 2 is a weaker pull-up or a write-assist (negative-BL or WL boost) in the column circuits rather than a cell change, so that the Part 1 characterization stays valid. For Part 2 we will keep this cell, mirror it to share BL contacts and the VDD/VSS rails, and size the wordline driver and precharge from $16 \times C_{BL}$ and $4 \times C_{WL}$ plus wire.

Traceability and team contributions

The handout used for this report is `docs/ese5700-project1.pdf` (Part 1 on pp. 3–4). Reproducible results are in `results/part1/read-snm/beta1/`, `beta1p5/`, and `beta2/`; each contains `raw.csv`, normalized curves and `snm.json`. The overlay coordinates are the two `lobes` entries in those JSON files. Recompute with the command below. The original handwritten and Cadence images are retained in `media/part1/`.

```
python3 scripts/check_snm_sweep.py -all
```

The selected design was also captured in separate Q- and $\overline{Q}$-side sweep schematics (Figure 17). These establish the nominally symmetric device assignment used by the two transfer-curve benches; they do not provide layout evidence.

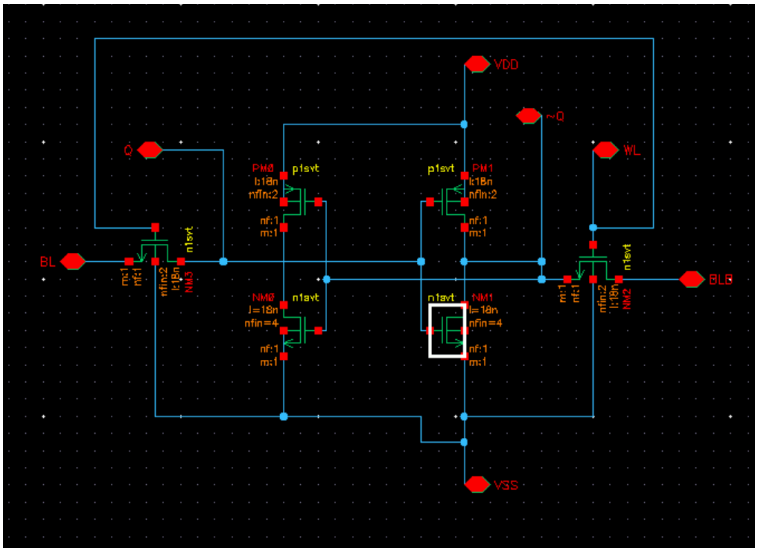


Figure 17: Original Cadence Q-side cell for $\beta = 2$.

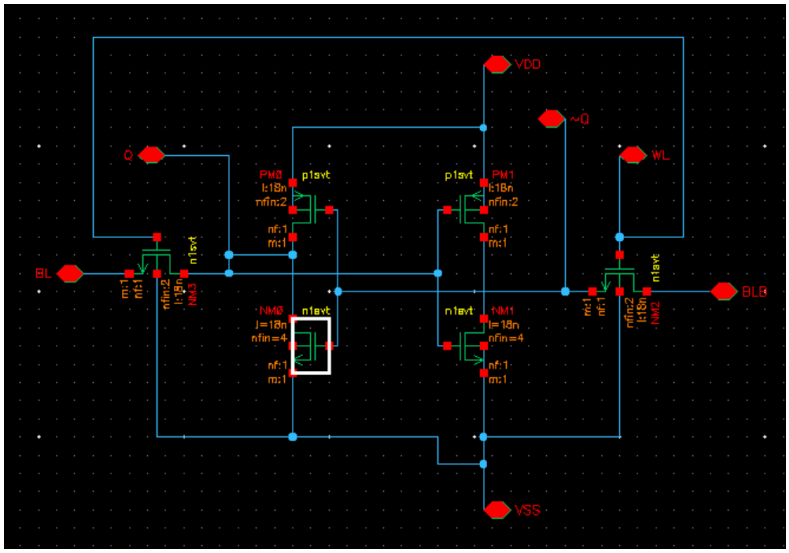


Figure 18: Original Cadence $\overline{Q}$-side cell for $\beta = 2$. The other PD variants remain in `media/part1/`.

Table 7: Team contributions (handout requirement). Both authors are responsible for understanding the complete design and report.

Tyrone Marhguy	**Not yet evidenced**
Victor Wanjohi	**Not yet evidenced**

The repository does not record which author performed each task; the two rows above must be filled by the team before upload.